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10 January 1969
Superseding
MIL-C-46982A (MU)
7 October 1964

MILITARY SPECIFICATION

CARTRIDGE, 4.2 INCH, HE, M329A1
W/SUPPLEMENTARY CHARGE, W/FUZE
PD, M557, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of high explosive cartridge designated as Cartridge, 4.2 Inch, HE, M329A1, W/Fuze, PD, M557.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-453 - Inspection, Radiographic
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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DRAWINGS

ARMY

7549247 - Container, Ammunition, Fiber, M251A1 for 4.2 Inch Mortar Ammunition in Jungle Wrap.
7549248 - Box, Packing Ammunition for 4.2 Inch Mortar Ammunition in Fiber Container, M251 in Jungle Wrap.
8863685 - Cartridge, 4.2 Inch, HE, M329A1, W/Supplemental Charge, for Mortar M2 and M30 Assembly.

PUBLICATIONS

ARMY

ET-8863685 - Equipment Tabulation

(Copies of specification, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material

3.1.1 General-Material and parts shall be in accordance with applicable specifications and drawings.

3.1.2 Clean Explosive Scrap.-Clean scrap (see 6.7) produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweeping, ventilation, and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used.

3.2 Loaded Cartridge.-The loaded cartridge shall comply with all requirements specified on drawing (dwg) 8863685, as applicable, and with all requirements specified in applicable specifications.

3.3 Cavitation in high explosive charge.

3.3.1 Projected areas and dimensions.-Cavities and cracks within the explosive charge shall not exceed the requirements specified in Table I when determined in accordance with 4.3.1.

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Cavities having a maximum (max) projected length of 1/32 inch or less shall be disregarded provided they are separated by 1/2 inch of solid cast.

TABLE I

	Segments (see Figure 1)			
	A	B	C	D
Sum of projected area of the cavities, excluding pipes, cracks and annular rings, square inch (sq. in.)	*1/64	1/4	1/2	1/2
Projected length of any cavity excluding pipes, cracks and annular rings (in)	1/8	1/2	1/2	3/4
Piping cavities, max. projected area (sq. in.)	0	1/4	1/2	Na.
Piping cavities, max. projected width (in)	0	1/4	1/4	Na.
Cracks, max. projected width (in)	1/32	1/32	1/32	Na.
Annular rings, max. projected width (in)	0	0	1/4	Na.

* If the length of the largest cavity is 1/16 inch or less, the max. total projected area may be 1/20 square inch.
NA - Not applicable

3.3.1.1 Porous areas.-Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.1.2 Cracked charges.-Not more than one transverse crack shall be permitted in Segment A and not more than a total of four transverse cracks shall be permitted in Segments A, B and C, combined. Cracks shall be disregarded in Segment D.

3.3.1.3 Annular rings.-The presence of annular rings 1/16 inch max. and extending from the shell wall 1/4 inch max. into the charge shall not be cause for rejection. The cavity area

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represented by such annular rings shall not be included when determining cavity area in the charge.

3.3.1.4 Cavities extending into two segments.-The entire area of cavities extending into two segments shall meet the requirements of both segments.

3.3.2 Surface cavitation.-Cavities formed in segment D of Figure 1 due to chipped or broken explosive from the side-walls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. The max. dimension of any visible cavity in the faced surface of the charge shall not exceed 1/4 inch in any direction and the total area of all cavities shall not exceed 80 percent of the surface area. No individual point of explosive shall extend above the flat surface on which the liner would rest. (Cavities having a max. dimension of 1/32 inch or less shall be disregarded in this computation). Disregard any cavitation in fuze well above Segement D.

3.4 Proving ground test.-The cartridge shall comply with the following requirements when tested as specified in 4.3.2.

3.4.1 Functioning

3.4.1.1 Failure to fire.-The cartridge shall not fail to fire.

3.4.1.2 Impact functioning.-The cartridge shall function high order on impact.

3.4.1.3 Horizontal range

3.4.1.3.1 No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.4.1.3.2 The standard deviation in horizontal range shall be no greater than the max. lot values shown in Table II for the increment charge indicated.

TABLE II

Increment Charge	Max. Lot Standard Deviation (meters)
7	26
41	72

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3.4.1.4 Premature burst.-There shall be no premature burst.

3.4.2 Safety.-There shall be no premature burst or indication of metal parts separation in the gun bore or in flight when fired at any service charge or with propellant adjusted to give a pressure of 112 precent of the rated max. pressure.

3.5 Workmanship

3.5.1 Cartridge.-All manufacturing, processing and assembly operations of the cartridge shall be correctly performed in a thorough workmanlike manner. Parts shall be free of burrs, chips, dirt, grease, rust, corrosion products and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.5.1.1 Loading of high explosive.-The interior surface of the projectile shall be dry when the high explosive is poured into the projectile. Solid explosives shall not be present in the projectile before loading or be added to the projectile during pouring or cooling.

3.5.1.2 Threads.-The mating surfaces of the threads shall be free of caked explosive and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care should be taken to prevent the chemcial cleaner from coming into contact with the charge and becoming entrapped in the projectile.

3.5.1.3 Exterior surface.-After loading, the exterior of the cartridge shall be clean and free of explosive. The painting and marking shall be restored by the loading and assembling contractor on any component furnished to him painted or marked when such painting or marking has been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications

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and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Metal parts or metal parts assemblies of one lot interfix number, from one manufacturer.
b. Loaded fuzes of one lot interfix number, from one manufacturer.
c. Loaded ignition cartridges, of one lot, from one manufacturer.
d. Supplementary charges of one lot interfix number from one manufacturer.
e. Propellant increments of one lot from one manufacturer.
f. Bag loading assemblies from one lot.
g. TNT of one lot interfix number from one manufacturer.
h. Loaded projectiles of one weight zone.

4.2.2 Examination.-Inspection for Critical defects (and Major defects, when so specified), shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Projectile loading assembly, prior to loading (see dwg. 8863929 covering a detail of dwg. 8863685)

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Categories	Defect	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Foreign matter in charge cavity (see 3.5)	Visual	01001
Major:	AQL 0.40 percent		
101.	Interior protective coating with bare spot	Visual	01002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	01003

4.2.2.2 Projectile loading assembly, prior to assembly of liner (see dwg. 8863929 covering a detail of dwg. 8863685).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Depth of fuze well, max.	Gage	02001
102.	Foreign matter in fuze well. . .	Visual	02003
Minor:	AQL 1.5 percent		
201.	Diameter of fuze well, max . . .	Gage	02004
202.	Surface cavitation excessive (see 3.3.2).	Visual	02005
203.	Evidence of poor workmanship (see 3.5).	Visual	02006
204.	Liner's External surface not coated, 360 degrees for entire length with silicone soft film .	Visual	02007

4.2.2.3 Projectile loading assembly, prior to assembling supplementary charge (see dwg. 8863929 covering a detail of 8863685).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		

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101.	Concentricity of inside diameter of liner with pitch diameter of thread (including perpendicularity with face of body)	Gage	03005
102.	Depth of bottom of liner, min. . . .	Gage	03006

Minor:	AQL 1.5 percent		
201.	Liner removable by hand.	Manual	03001
202.	Liner missing or damaged	Visual	03002
203.	Foreign matter in supplementary charge cavity.	Visual	03003
204.	Evidence of poor workmanship (see 3.5).	Visual	03004

4.2.2.4 Supplementary charge prior to assembly into projectile loading assembly (see dwg. 8797090 a detail of dwg. 8863685).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Supplementary charge pad missing . .	Visual	04001
Major:	None defined.		
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	04002

4.2.2.5 Projectile loading assembly, prior to assembling fuze and obturating mechanism (see dwg. 8863685).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Supplementary charge missing or inverted	Visual	05001
102.	Supplementary charge nor removable by hand or tab missing.	Visual-Manual	05002

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Minor:	AQL 1.0 percent		
201.	Supplementary charge damaged . . .	Visual	05003
202.	Evidence of poor workmanship (see 3.5).	Visual	05004

4.2.2.6 Cartridge assembly, prior to assembling pressure plate (see dwg. 8863685).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	100 percent		
101.	Rotating disc and cartridge container fail to withstand required disassembly torque. . . .	Test	06001
Major:	0.40 percent		
102.	Rotating disc not properly assembled.	Visual	06002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5).	Visual	06003

4.2.2.7 Cartridge assembly, after assembly of pressure plate and pressure plate nut (see 8863685).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Pressure plate incorrectly assembled (reversed)	Template	07001
2.	Pressure plate assembled so red face is visible	Visual	07004
Major:	AQL 0.40 percent		
101.	Pressure plate nut not properly staked.	Visual	07002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5).	Visual	07003

4.2.2.8 Cartridge assembly, after assembling ignition cartridge and striker nut assembly (see dwg. 8863685)

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Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Extension pin not free to slide in and bottom in extension	Visual-Manual	08007
Major:	AQL 0.65 percent		
101.	Striker nut fails to withstand required disassembly torque . . .	Test	08001
102.	Striker point missing or loose. .	Visual-Manual	08002
103.	Ignition cartridge missing. . . .	Visual	08003

Minor: None defined.

4.2.2.9 Cartridge assembly, prior to packing (see dwg. 8863685).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Incorrect number of increments. . .	Visual	09001
2.	Warning label missing or unidentifiable.	Visual	09002
3.	Bag loading assembly missing, cut, torn, or damaged to the extent that propellant will escape.	Visual	09020
Major:	100 percent inspection		
101.	Extension pin missing	Visual	09005
Major:	AQL 0.65 percent		
102.	Fuze fails to withstand required disassembly torque.	Test	09003
103.	Diameter of bourrelet, max. (if retouched).	Gage	09004
104.	Any propellant holder missing . .	Visual	09007
105.	Any component damaged or improperly assembled.	Visual-Manual	09008
Minor:	AQL 1.5 percent		
201.	Fuze not staked	Visual	09009
202.	Marking misleading or unidentifiable including incorrect color code.	Visual	09010

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203.	Protective coating with total damaged area in excess of 1/4 square inch . .	Visual	09011
204.	Evidence of poor workmanship (see 3.5)	Visual	09012

4.2.2.10 Unsealed fiber container (see dwg. 7549247)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Spacer tube missing	Visual	10001
102.	Support tube missing	Visual	10002
Minor:	AQL 1.5 percent		
201.	Glue or asphalt on round, . .	Viusal	10003
202.	Cartridge improperly assembled into container. . .	Visual	10004
203.	Round cannot be removed by hand.	Visual	10005

4.2.2.11 Sealed fiber container (see dwg. 7549247)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Container damaged or cut through all asphalt impregnated layers.	Visual	11001
102.	Tape incomplete or badly wrinkled.	Visual	11002
103.	Metal end loose or distorted.	Visual-Manual	11003
Minor:	AQL 1.5 percent		
201.	Tear tab length improper. . .	Visual-Gage	11004
202.	Gap between cover and body of container exceeds 1/8 inch.	Visual	11005
203.	Marking misleading or unidentifiable.	Visual	11006
204.	Contents loose.	Manual	11007

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4.2.2.12 Fiber container wrapped in barrier material prior to waxing - Jungle Wrap (see dwg. 7549247).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Barrier material improperly assembled	Visual	16001
102.	Barrier material ends improperly folded or taped	Visual	16002
Minor:	AQL 1.0 percent		
201.	Marking missing or incomplete . . .	Visual	16003

4.2.2.13 Fiber container after waxing - Jungle Wrap (see dwg. 7549247).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Wax coating incomplete or damaged	Visual	17001
Minor:	AQL 1.0 percent		
201.	Marking missing or unidentifiable	Visual	17002
202.	Kraft paper improperly placed on container	Visual	17003

4.2.2.14 Sealed wood packing box (see dwg. 7549248).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.65 percent		
101.	Contents exposed	Visual	12001
102.	DOD symbol missing or incorrect.	Visual	12002
103.	Board broken or split	Visual	12003
104.	Hardware or strapping missing, broken or loose.	Visual-Manual	12004
Minor:	AQL 2.5 percent		
201.	Marking misleading or unidentifiable	Visual	12005
202.	Hardware improperly engaged.	Visual-Manual	12006
203.	Car seal missing, unsealed or improperly positioned.	Visual-Manual	12007
204.	Contents loose	Visual-Manual	12008
205.	Handle missing or insecure	Visual-Manual	12009

4.2.3 Testing

4.2.3.1 Check test for possible deterioration of ignition cartridge Code No. 08004.-If the total elapsed time between the original acceptance of the ignition cartridge lot and the assembly of that lot into complete round cartridges exceeds two years, or if the ignition cartridges have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to, and must satisfactorily pass, the check test for deterioration specified in the applicable ignition cartridge loading specification immediately before the cartridge lot is assembled into the complete round cartridge. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (see 6.4).

4.2.3.2 Cavitation in high explosive charge (see 3.3 except 3.3.2) Critical defect - Code No. 02007.-The sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-1235, using a CSP-1 plan having an AQL of 0.65 percent. Appropriate i and f values shall be as specified by the procuring

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activity. Any cartridge having cavitation in excess of that permitted by the applicable requirement for any segment shall be classed defective. The test shall be performed as specified in 4.3.1, using equipment in accordance with 4.2.4.

4.2.3.3 Profile and alignment (see dwg. 8863685) Major defect-Code No. 09013.-The cartridges shall be gaged 100 percent for profile and alignment. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.3.1 Torque test (dwg. 8863685, note 4).

4.2.3.3.1.1 Twenty-thirty inch-pound disassembly torque - Major defect - Code No. 08006.-The cartridge shall be tested 100 percent for this requirement. Any cartridges which fail to meet the requirement shall be classed defective and removed from the lot or reworked to meet the requirement.

4.2.3.3.1.2 Fifty inch-pound disassembly torque-Major defect-Code No. 08005.-A sample of fifty cartridges shall be tested for the requirement specified. If two or more cartridges fail the requirement, the lot shall be rejected and reworked by loosening the cartridge container extension to the degree necessary to meet the fifty inch-pound disassembly torque.

4.2.3.4 Projectile zone weight (see dwg. 8863929) Major defect - Code No. 03005.-The projectile shall be weighed and inspected for correct weight zone marking 100 percent. Any projectile which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.5 Proving ground tests.-The cartridges shall be observed for compliance with the requirements as specified in 4.2.3.5.1.

4.2.3.5.1 Functioning.-Failure to fire (see 3.4.1.1) Major defect, Code No. 09014; impact functioning (see3.4.1.2) Major defect, Code No. 09015; range standard deviation (see 3.4.1.3.2) Major defect, Code No. 09016; premature burst (see 3.4.1.4) Critical defect, Code No. 09017 and range less than 80 percent average range (see 3.4.1.3.1) Critical defect, Code No. 09019.

4.2.3.5.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, 80 sample cartridges shall be selected from each lot for this test. If two or more failures

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to fire occur, if four or more impact functioning failures occur or if a critical defect occurs, the lot shall be rejected. In addition, if the standard deviation in horizontal range multiplied by .79 exceeds the requirement specified in Table II, the lot shall be rejected. The test shall be performed as specified in 4.3.2.

4.2.3.5.1.2 After three consecutive lots have complied with the criteria of 4.2.3.5.1.1, 50 sample cartridges shall be selected from each lot for this test. If two or more failures to fire occur, if three or more impact functioning failures occur or if a critical defect occurs, the lot shall be rejected. In addition, if the standard deviation in horizontal range multiplied by 0.75 exceeds the requirement specified in Table II, the lot shall be rejected. The test shall be performed as specified in 4.3.2.

4.2.3.5.2 Safety (see 3.4.2) Critical defect, Code No. 09018.

4.2.3.5.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirement specified, 20 sample cartridges shall be selected at random from each lot for this test. If any cartridge fails to comply with the safety requirements, the lot shall be rejected. The test shall be performed as specified in 4.3.2.

4.2.3.5.2.2 After three consecutive lots have met the criteria of 4.2.3.5.2.1, the sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter E (single sampling) utilizing an AQL of 1.0 percent. If any cartridge fails to comply with the safety requirements, the lot shall be rejected. The test shall be performed as specified in 4.3.2.

4.2.4 Inspection Equipment-Equipment tabulation Number ET-8863685 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

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4.2.4.3 Government use of contractor's inspection equipment.- The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Determination of cavities.-The sample cartridges shall be subjected to x-ray examination. In all exposures the x-ray beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination of the x-ray negative shall be made to determine compliance with applicable requirements. Radiographic equipment, operations and procedures shall be qualified in accordance with Specification MIL-STD-453. In addition, the sensitivity of x-ray technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25, as determined in accordance with 4.3.1.2. In the event an original x-ray picture leaves doubt as to acceptability, the cartridge shall be re-x-rayed at an angle 90 degrees from that of the original exposure and in the same plane.

4.3.1.1 Check of sensitivity of x-ray technique.-The sensitivity of x-ray technique shall be determined, by the means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the projectile away from the film (source side). Observation shall be made of the image of the penetrameter on the x-ray film. The sensitivity of x-ray technique shall be considered satisfactory when the image of the penetrameter including the 2T hole in the center, is sharply defined on the x-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thicknesses and shall be equal to 2.0 percent of the max. equivalent thickness. The penetrameter shall have a hole the diameter of which is twice the thickness of the penetrameter. For further details on the construction of the penetrameter see MIL-STD-453. The actual thickness may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated by Figure 2 of this specification.

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4.3.1.2 Check of photographic density of film.-The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to that portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards be used (see 6.6) comparison of film with standards may be made visually; however the use of a densitometer is recommended.

4.3.2 Proving ground test.

4.3.2.1 Functioning.-The cartridges for this test shall be divided into two equal groups with ten of the fuzes in each group set for delay functioning and the balance set for super-quick functioning. Group I shall be fired with 7 increments of propellant and the mortar elevation shall be 1065 mils. Group II shall be fired with 41 increments of propellant and the mortar elevation shall be 900 mils or below. The cartridges shall be fired from a mortar for which the rounds are standard. For comparison, to make obvious the introduction of any variables that would influence round performance, reference rounds shall be fired with each test (no more than three test lots with one reference lot). The individual velocities, and impact ranges sha. be obtained and recorded. The standard deviation and the arithmetic average of the impact ranges and velocities shall be calculated and recorded. The individual maximum pressures shall be obtained and recorded for informational purposes for the first three lots. Hawk radar shall be used to observe for metal parts security and in the case of short rounds or unobserved dud rounds, radar magnetic tape records of such rounds along with the three preceding and three following rounds shall be forwarded for analysis to Commanding Officer, Picatinny Arsenal, Dover, New Jersey ATTN: SMUPA-ND7.

4.3.2.2 Safety.-The cartridges for this test shall be assembled with the propellant adjusted to give a pressure of 112 percent (17,500 psi) of the rated max. pressure with the fuzes set super-quick, and fired from a mortar with the elevation at 900 mils. Observation shall be made for premature bursts and indication of metal parts separation in the mortar bore or in flight.

4.3.2.3 Refiring.-If for any reason the proving grounds or the technical agency considers that the test conditions have detrimentally affected the test results, the test results shall be declared invalid and a new test shall be performed on additional

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sample cartridges as required. Reference rounds shall be fired along with retest lots to make obvious the introduction of any variables that would influence round performance. Only one lot of reference rounds shall be fired each day.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with dwg. 7549247.

5.2 Packing

5.2.1 Level A.-The containers shall be packed in accordance with dwg. 7549248.

5.3 Marking.-The containers shall be marked in accordance with dwg. 7549247, and the wood crates shall be marked in accordance with dwg. 7549248.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor forma and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detaile drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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6.4 Cost of check test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Submission of test data.-In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data will be forwarded to the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

6.6 Photographic densities.-"American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, New York, has been found staisfactory as a source of information on filter densities.

6.7 Clean scrap.-Clean scrap is defined as scrap formed in kettles, pails, risers, and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation.

6.8 Supersession data.-This specification supersedes MIL-C-46982A(MU) dated 7 October 1964 covering Cartridge, 4.2 Inch, HE, M329A1 w/supplementary charge, w/fuze PD, M557, Loading Assembling and Packing.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1315-A-672

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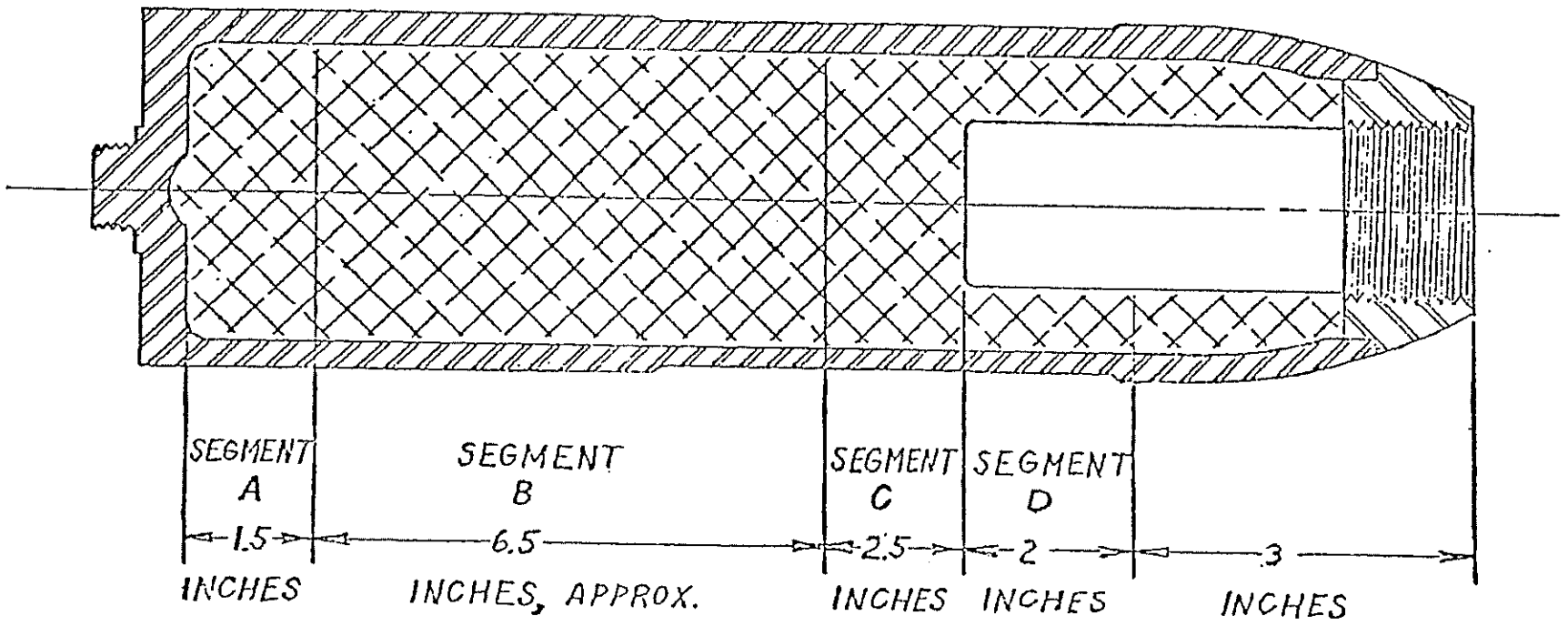
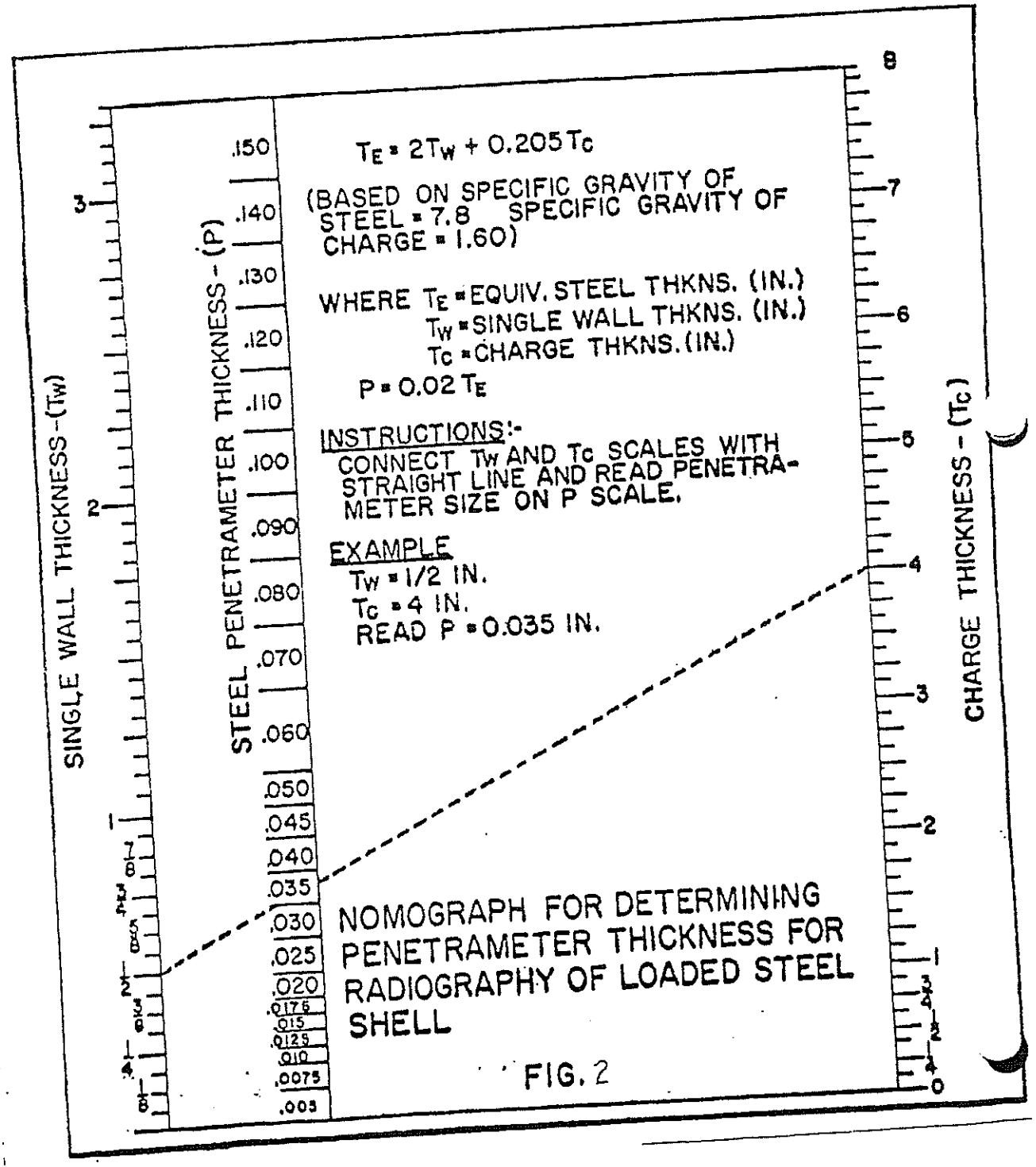


FIGURE I

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NOMOGRAPH FOR DETERMINING PENETRAMETER THICKNESS FOR RADIOGRAPHY OF LOADED STEEL SHELL

FIG. 2

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